Term Project

Project

- Database Design: ER/EER Diagram, Relational Mapping
- Objective: **Develop a mini web application with a focus on e-commerce functionality.** Your project will involve designing an ER diagram for the database and implementing a functional web app

Presentation: 10-15 mins (**idea, database design, app demo**)

- TBA

Project

Web Application Features

🔑 Core Features

- **Home Page** – general landing page for the site.
- **Products Page (public)** – visitors can see available products.
- **Register Page** – allow new customers to sign up.
- **Log-in Page** – members must log in before placing an order.
- **Member Info Page** – profile page showing member details.
- **Cart Page** – members can add products to cart.
- **Summary / Checkout Page** – review items and confirm order.

Project

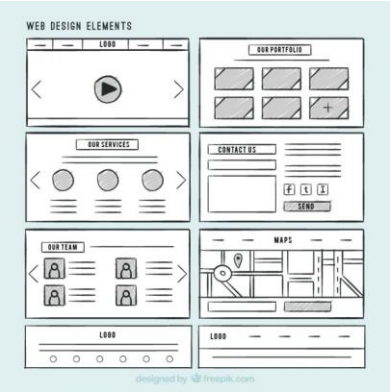
You must implement **at least three** of the following:

🔑 Extra Features

- **Promotion (Discount) such as coupons:** Apply % discount to selected products.
- **Promotion (Buy X Get Y):** e.g., buy 2 get 1 free.
- **Member Royalty Program:** Membership levels (such as Silver, Gold, Platinum) with benefits.
- **Wish List Page:** Customers can add products to a personal wish list.
- **Comments / Reviews:** Customers can post textual reviews on products.
- **Rating System:** Customers rate products (1–5 stars).
- **Admin Dashboard:** manage users, products, and orders.
- **Sales Report**

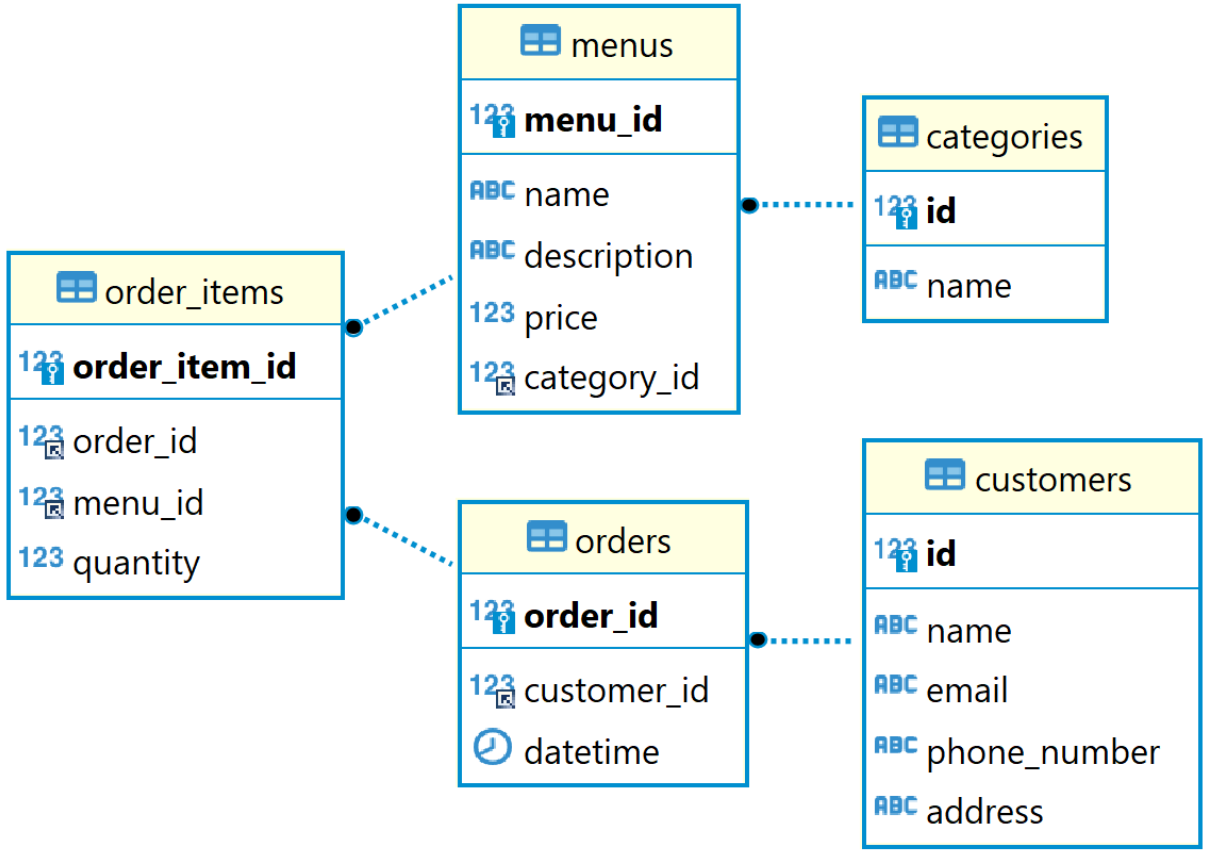
The files that you need to submit are as follow

1. The database schema (ER, EER, UML diagram)
2. Relational Mapping
3. Low-fidelity prototype



4. Presentation file
5. Source code

An Example of Database Schema

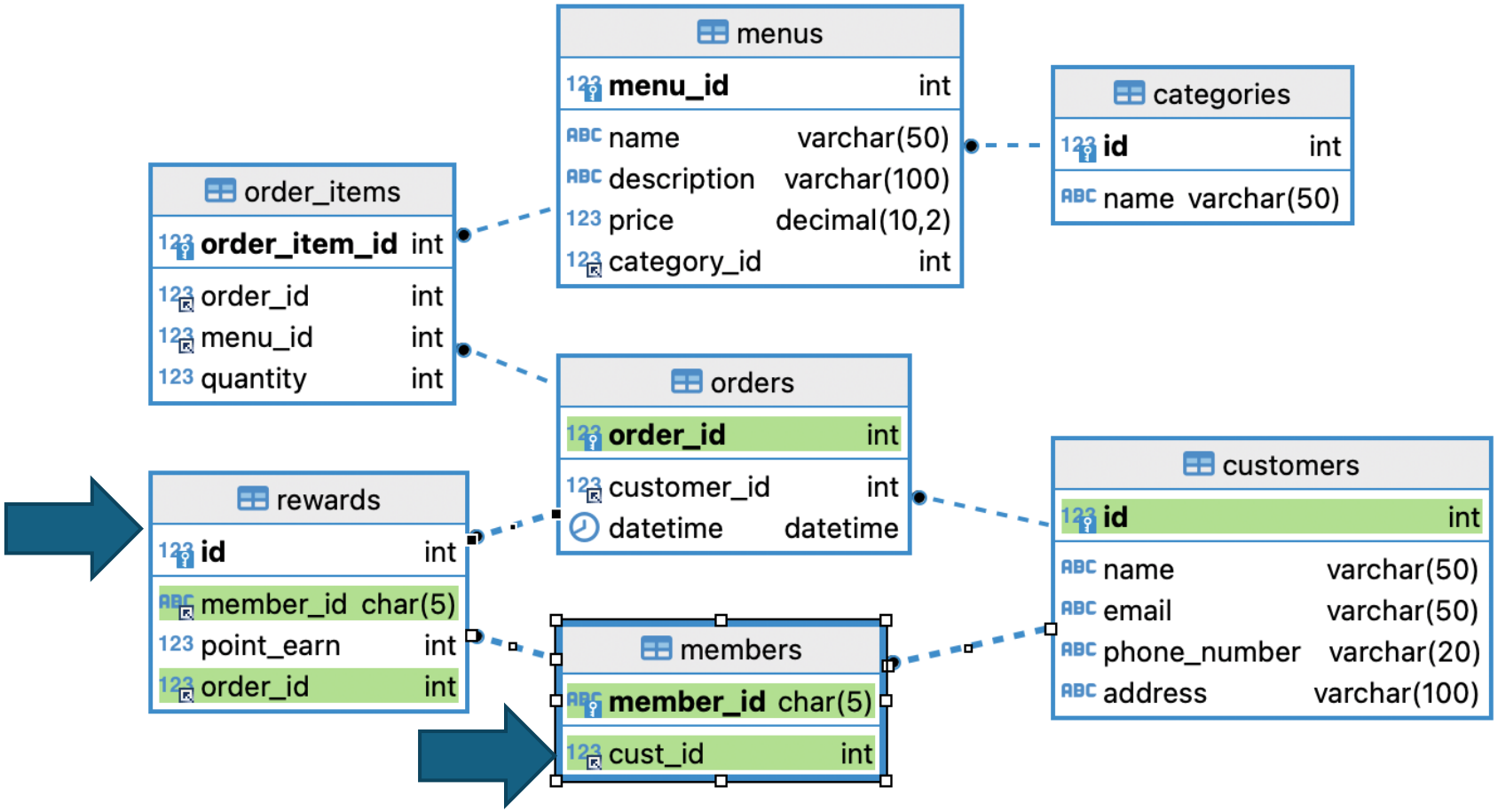


An example of modified schema

Suppose that we would like to include a reward program for our store.

Do we need to modify the database schema?

Which information that we need to store?



Presentation (example)

- Introduction
  - Ideas
  - Mock-up
- Relational Model
  - Database Design (ER, EER, UML, Mapping)
- App Demo (on presentation date)
  - Demonstrate the functionality of the web application.